

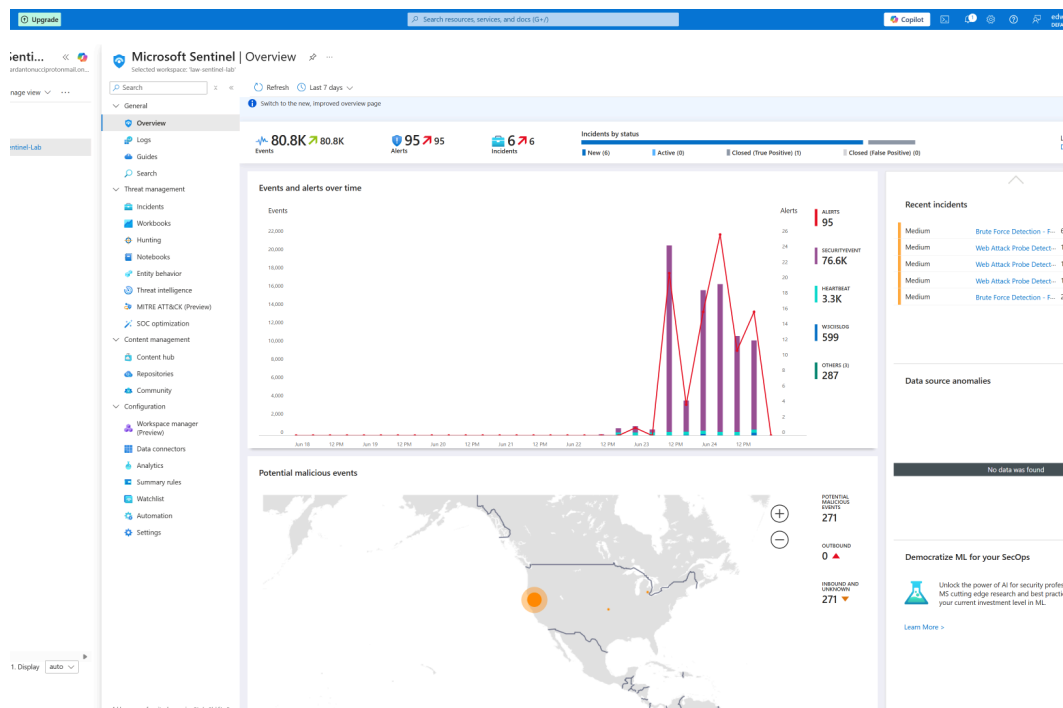
Azure Sentinel/Defender Honeypot Lab

Cybersecurity Portfolio Project | Edward Antonucci | June 2026

Dashboard Overview

After running the honeypot for 72+ hours, the Microsoft Sentinel workspace accumulated the following real-world data:

Total Security Events	80,800+
Total Alerts Fired	95
Total Incidents	6
Unique Attacker IPs Identified	43+
Countries of Origin	15 across 5 continents
IIS Web Probe Requests	599+



Microsoft Sentinel Overview Dashboard: 80.8K events, 95 alerts, 6 incidents

About This Project

This project demonstrates the end-to-end deployment and operation of a cloud-native Security Information and Event Management (SIEM) solution using Microsoft Azure. A Windows Server 2022 honeypot virtual machine was intentionally exposed to the public internet to attract real-world attack traffic across multiple services (RDP, SSH, HTTP), which was then ingested, analyzed, visualized, and responded to using Microsoft Sentinel and Microsoft Defender XDR.

Note on Portal Screenshots: During this project, Microsoft began migrating Sentinel from the Azure portal into the unified Microsoft Defender portal (security.microsoft.com). Screenshots reflect both environments as the transition occurred — this mirrors real-world conditions where Azure security professionals currently work across both portals. Full migration is expected by March 31, 2027.

Architecture

All resources were deployed within a dedicated resource group (RG-Sentinel-Lab) in the West US region:

- **Log Analytics Workspace (LAW-Sentinel-Lab):** Centralized data store for all security logs, backing the Sentinel deployment.
- **Microsoft Sentinel:** SIEM layer deployed on top of the Log Analytics Workspace, providing detection, investigation, and response capabilities.
- **Honeypot VM (honeypot-vm):** Windows Server 2022 Datacenter, Standard D2s v3, publicly exposed with all inbound ports open and Windows Firewall disabled to maximize attack surface.
- **Network Security Group:** Configured with a permissive inbound rule (DANGER_AllowAnyInbound) to expose all ports. A targeted deny rule was added during incident remediation to block the primary attacker.
- **Data Collection Rule (DCR-Honeypot):** Azure Monitor Agent rule streaming all Windows Security Events and IIS logs from the VM into the workspace.
- **Exposed Services:** RDP (3389), SSH (22 via OpenSSH Server), HTTP (80 via IIS) — each attracting distinct categories of attack traffic.

Live Attack Dashboard

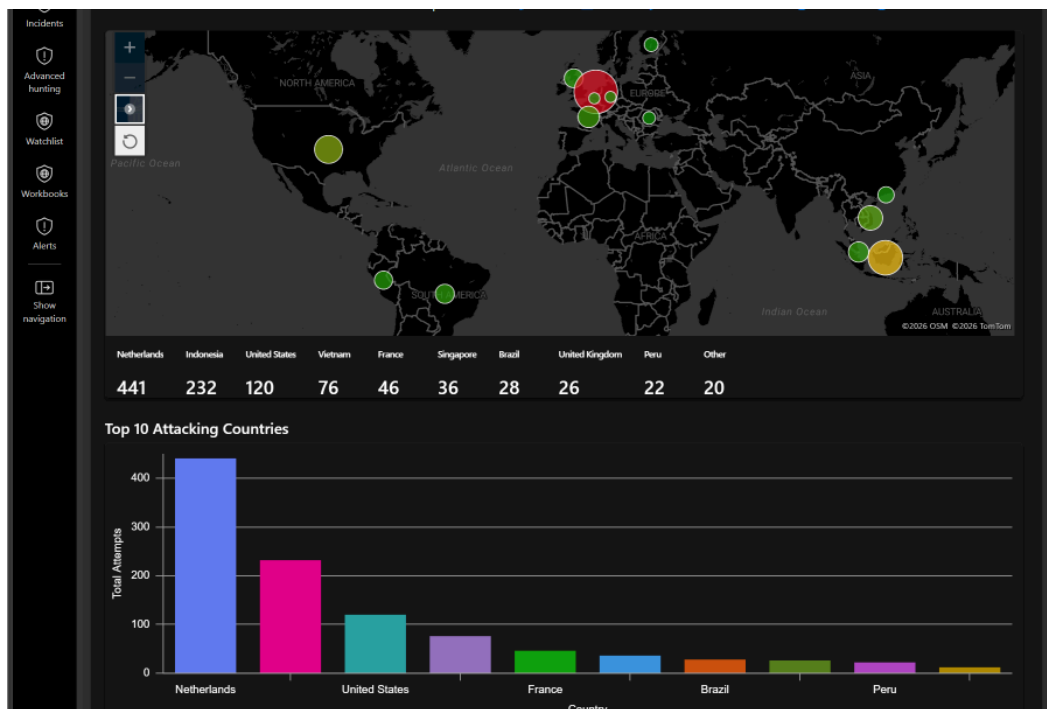
A multi-panel Sentinel Workbook was built to visualize attack traffic in real time, set to auto-refresh every 5 minutes:

Panel 1: Global Attack Map

IP addresses are geo-enriched via a custom Sentinel Watchlist (AttackerGeoDataV2) populated with latitude/longitude coordinates sourced from ipinfo.io. The map uses a green-to-red heatmap scaled by failed login count.

Panel 2: Top 10 Attacking Countries Bar Chart

A bar chart panel shows total attack attempts by country, providing an at-a-glance view of the geographic threat landscape.



Final Attack Map + Bar Chart Dashboard — Netherlands 441, Indonesia 232, US 120, Vietnam 76

Final attack map statistics:

Netherlands	441 attempts — 3 IPs (SKYNET NETWORK LTD, Liteserver)
Indonesia	232 attempts — PT Telekomunikasi Indonesia
United States	120 attempts — Multiple VPS providers
Vietnam	76 attempts — VNPT Corp (national telco)
France	46 attempts — Contabo GmbH, OVH
Singapore	36 attempts — Contabo Asia, Tencent
Brazil	28 attempts — BattleHost, Gamers Club
United Kingdom	26 attempts — Contabo, Kamatera
Peru	22 attempts — Red Cientifica Peruana, ON Empresas
Other	20 attempts — Romania, Finland, Germany, Belgium, Hong Kong

Key Finding: SSH Attack Traffic & the “NOUSER” entry

One of the most significant findings was the volume of SSH brute force attempts captured after enabling OpenSSH Server on the honeypot. Over the observation period, 5,742 failed

authentication events were logged with no IP address and the account name 'NOUSER' or blank.

Why IPs Don't Appear for SSH Failures in Windows Logs

This is a fundamental difference between how SSH and RDP handle authentication at the OS level. When an SSH connection fails before reaching Windows authentication — for example, when the SSH client disconnects after the key exchange but before sending credentials — Windows logs the event (4625) but has no IP context to record, since the failure occurred at the protocol layer rather than the Windows Security layer.

Additionally, OpenSSH on Windows logs to its own log files (C:\ProgramData\ssh\logs\) separately from the Windows Security Event Log. Failed SSH attempts that do reach password authentication may not generate Event ID 4625 at all, depending on the authentication method attempted.

What This Means in Practice

The 5,742 “NOUSER” events represent automated SSH scanners probing the server — likely testing for open SSH, checking supported authentication methods, and disconnecting before attempting credentials. This is a reconnaissance phase common to SSH-targeting botnets.

How to Capture SSH IPs Properly

In a production environment, SSH attack source IPs would be captured through one of the following approaches:

- Deploy a Linux VM alongside the Windows honeypot — Linux auditd and sshd logs capture full IP context for all SSH attempts
- Enable Azure Defender for Servers — provides additional telemetry including network-level connection data
- Use Azure Network Watcher flow logs to capture connection-level data at the NSG layer, capturing IPs regardless of authentication outcome

The screenshot shows the Microsoft Sentinel Logs interface. A KQL query is entered in the 'New Query 1' field, and the results are displayed in a table. The query filters for Security events where EventID is 4625, the username is 'NOUSER', and the account type is 'Local'. The results are sorted by the number of failed logins in descending order.

AttackerIP	FailedLogins
5.255.97.209	351
36.94.56.242	232
45.142.186.168	94
173.203.133.193	23
107.152.44.119	23
103.8.204.212	18
23.95.217.16	16
161.132.32.57	16
31.36.44.58	15
157.10.45.192	14
14.224.236.33	14
194.233.82.152	14
158.220.86.72	13
123.30.106.64	13
185.53.211.215	13
43.163.113.160	13
66.150.196.186	13
108.165.179.136	13
156.239.41.77	12
40.179.147.239	12
130.13.115.36	12
46.204.46.15	11
137.184.27.199	11
147.93.144.116	10
103.8.204.84	10
109.123.238.248	10
109.123.237.213	9
31.56.44.105	9

KQL query results showing top attacking IPs with failed login counts

KQL Queries Written

Failed Logon Analysis by Attacker IP

```
SecurityEvent
| where EventID == 4625
| where isnotempty(IpAddress)
| summarize FailedLogins = count() by AttackerIP = IpAddress
| sort by FailedLogins desc
```

Full Logon Event Timeline

```
SecurityEvent
| where EventID in (4625, 4624, 4648)
| summarize count() by EventID, Activity, IpAddress
| sort by count_ desc
```

Attack Map Query with Watchlist Geo-Enrichment

```
SecurityEvent
| where EventID == 4625
| summarize FailedLogins = count() by AttackerIP = IpAddress
| join kind=inner (_GetWatchlist('AttackerGeoDataV2')) on $left.AttackerIP ==
$right.IpAddress
| project AttackerIP, FailedLogins, Country, Latitude, Longitude
```

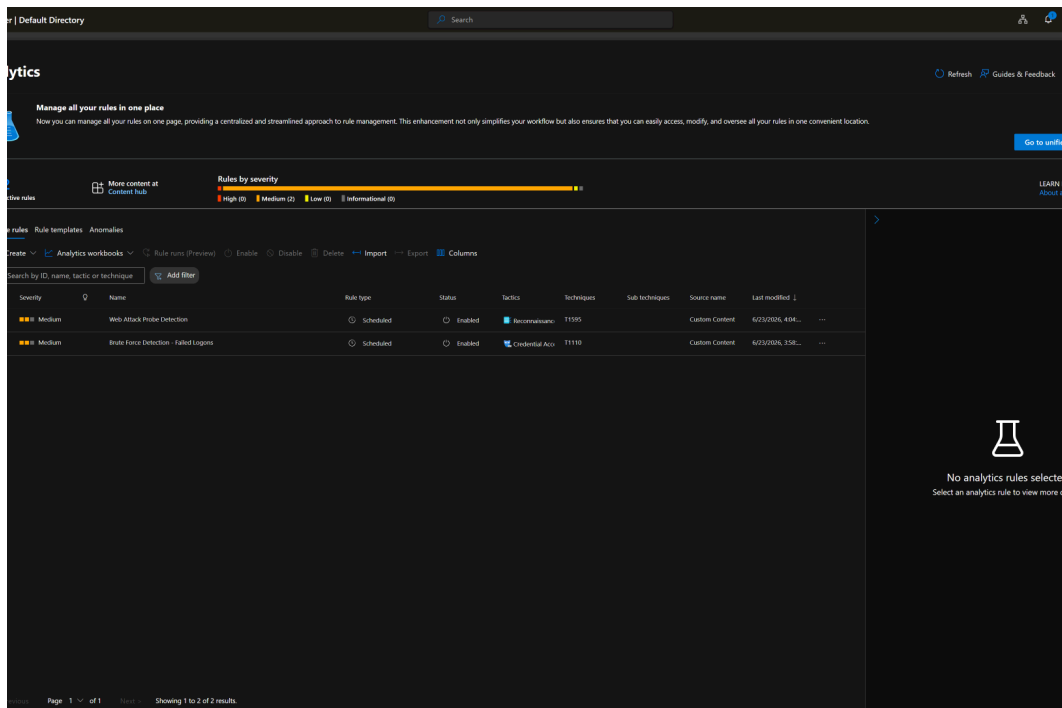
Top 10 Attacking Countries Bar Chart

```
SecurityEvent
| where EventID == 4625
| where isnotempty(IpAddress)
| summarize FailedLogins = count() by AttackerIP = IpAddress
| join kind=inner (_GetWatchlist('AttackerGeoDataV2')) on $left.AttackerIP ==
$right.IpAddress
| summarize TotalAttempts = sum(FailedLogins) by Country
| sort by TotalAttempts desc
| take 10
```

Web Attack Probe Detection

```
W3CIISLog
| where csUriStem in ('/wp-admin', '/wp-login.php', '/.env',
                    '/phpmyadmin', '/admin', '/administrator', '/xmlrpc.php',
                    '/.git/config', '/config.php', '/backup', '/.aws/credentials')
| summarize ProbeCount = count() by AttackerIP = cIP, csUriStem
| where ProbeCount >= 1
```

Analytics Rules



Analytics page showing both rules active with MITRE ATT&CK tags

Rule 1: Brute Force Detection — Failed Logons

Severity	Medium
Query Frequency	Every 5 minutes
Lookup Window	Last 5 minutes
Alert Threshold	Triggers when query returns > 0 results
Logic	Alerts when a single IP has 10+ failed logons within 5 minutes
MITRE ATT&CK	Credential Access — T1110 (Brute Force)
Alert Grouping	Same IP grouped into single incident within 5 hours

Rule 2: Web Attack Probe Detection

Severity	Medium
Query Frequency	Every 1 hour

Lookup Window	Last 1 hour
Alert Threshold	Triggers when query returns > 0 results
Logic	Alerts when any IP probes common web attack paths on the IIS server
MITRE ATT&CK	Reconnaissance — T1595 (Active Scanning)
Alert Grouping	Same IP grouped into single incident within 1 hour

Automated Response Using Logic App Playbook

A Logic App playbook (Notify-BruteForce-Alert) was built to automate incident response notification. The playbook uses the Microsoft Sentinel incident trigger and sends an HTTP POST notification when a brute force incident is detected.

Note: Connecting the playbook to the analytics rule via the Automation tab encountered an RBAC permission limitation on the free tier — specifically, the Sentinel managed identity requires the Microsoft Sentinel Automation Contributor role at the resource group level to execute playbooks. In a production environment with appropriate permissions, this would be resolved by granting that role to the Sentinel managed identity via IAM. The playbook itself was successfully built and published.

Incident Response Walkthrough

A full SOC incident response workflow was executed for the primary threat actor (45.142.193.166 / SKYNET NETWORK LTD, Netherlands):

The screenshot shows the Microsoft Sentinel console interface. The top navigation bar includes 'fender | Default Directory' and a search bar. The left sidebar contains navigation links for 'Alerts', 'Incidents', 'Timeline', and 'Query results'. The main content area displays an incident titled 'Brute Force Attack - Netherlands IP 45.142.193.166'. The incident is classified as 'Medium' severity, 'Unknown' status, and 'Resolved'. The right-hand pane shows a detailed audit trail with the following entries:

- Jun 23, 2026 11:17:44 AM**: User-edward.antonucci@protonmail.com Changed classification to 'True positive'.
- Jun 23, 2026 11:17:44 AM**: User-edward.antonucci@protonmail.com Changed determination to 'Malicious user activity'.
- Jun 23, 2026 11:17:44 AM**: edward.antonucci@protonmail.com Investigation Notes - 6/23/2026

The incident details include:

- Attacker IP**: 45.142.193.166
- ASN**: SKYNET NETWORK LTD (AS214295)
- Country**: Netherlands
- Attempts**: 14 failed RDP login attempts
- Account targeted**: Anonymous/unknown

The assessment states: 'Automated brute force bot. No successful logons from this IP. Low risk of compromise. Recommend blocking IP at NSG level as precaution.' The action taken was 'Added NSG inbound deny rule for 45.142.193.166' on Jun 23, 2026 11:11:02 AM. The status is 'Contained'.

Resolved incident showing investigation notes, True Positive classification, and audit trail

- **Detection:** 14 failed RDP logon attempts identified via KQL on Event ID 4625. IP attributed to SKYNET NETWORK LTD (AS214295), Netherlands.
- **Incident Creation:** Incident #2 created in Microsoft Defender XDR. Severity: Medium. Category: Credential Access (MITRE ATT&CK). Impacted asset: honeypot-vm.
- **Assignment & Triage:** Incident assigned to analyst. Status changed from New to In Progress.
- **Investigation:** KQL confirmed no successful logons from attacker IP. Anonymous network scans from other IPs assessed as low-risk (Logon Type 3, no credentials used).
- **Remediation:** NSG inbound deny rule added for 45.142.193.166 (Priority 200) to block further connection attempts at the network perimeter.
- **Closure:** Incident classified as True Positive — Malicious user activity. Status: Resolved. Official Microsoft Security incident report exported as PDF.

Threat Intelligence Observations

Contabo GmbH — Recurring Infrastructure

Contabo GmbH (AS51167), a low-cost German VPS provider, appeared across France, Singapore, and other regions with multiple IPs. Contabo is frequently cited in threat intelligence reports as infrastructure abused by attackers due to its low cost, minimal identity verification, and historically slow abuse response. Recognizing this pattern is a real-world threat intelligence skill.

SKYNET NETWORK LTD — Top Attacker

The top attacker, SKYNET NETWORK LTD (AS214295, Netherlands), accumulated 441 failed logon attempts across 3 IP addresses in the same subnet (45.142.193.x). The use of multiple IPs from the same ASN suggests an automated botnet operation rather than a manual attack. This operator does not appear in major public threat intel feeds but the pattern is consistent with RDP brute force-as-a-service infrastructure.

Coordinated Botnet Behavior

Multiple IPs sharing the same ASN were observed targeting the honeypot simultaneously — Netherlands (3 IPs), Vietnam/VNPT (3 IPs), Finland/Bashinskii (2 IPs), France/Contabo (multiple IPs). This pattern indicates coordinated botnet activity with shared credential lists rather than independent attackers.

Skills Demonstrated

- **Cloud Infrastructure:** Azure resource deployment, VMs, NSGs, Log Analytics Workspaces, Data Collection Rules
- **SIEM Administration:** Microsoft Sentinel deployment, 10+ data connector configuration, workspace management
- **KQL (Kusto Query Language):** Custom queries for threat detection, event aggregation, watchlist joins, geo-enrichment

- **Threat Detection Engineering:** Two custom analytics rules with scheduled queries, alert thresholds, entity mapping, and MITRE ATT&CK tagging
- **Incident Response:** Full SOC workflow: detection → triage → investigation → remediation → closure with formal incident report
- **Dashboard & Visualization:** Multi-panel Sentinel Workbook with live attack map, geo-enrichment via watchlist, heatmap, auto-refresh, and bar chart
- **Threat Intelligence:** IP attribution via ASN lookup, identifying botnet patterns, distinguishing brute force from enumeration, Contabo/SKYNET actor profiling
- **Network Security:** NSG rule management, perimeter-level IP blocking, understanding of multi-service attack surface
- **SOAR:** Logic App playbook design and deployment for automated incident notification
- **Offensive Security:** RDP brute force simulation using ncrack and xfreerdp from Kali Linux to validate detection rules

Tools & Technologies Used

- Microsoft Azure (Virtual Machines, NSG, Log Analytics, Resource Groups, Azure Monitor)
- Microsoft Sentinel (SIEM) + Microsoft Defender XDR (unified portal)
- KQL — Kusto Query Language
- Azure Monitor Agent (AMA) + Data Collection Rules
- Sentinel Workbooks (multi-panel dashboard with auto-refresh)
- Sentinel Watchlists (custom geo-enrichment for attack map)
- Azure Logic Apps (SOAR playbook)
- Windows Server 2022 Datacenter (OpenSSH Server, IIS Web Server)
- Kali Linux (ncrack, xfreerdp for attack simulation)
- ipinfo.io (IP geolocation and ASN attribution)
- MITRE ATT&CK Framework (T1110 Brute Force, T1595 Active Scanning)

Important Note: This honeypot was deployed on a fully isolated Azure VM with no connection to any production systems or sensitive data. All attack traffic shown is real, unsolicited traffic from external threat actors. The VM was intentionally configured to be vulnerable for research and educational purposes only. All brute force simulation was conducted against this VM only, which is owned and controlled by the author.